STATISTICS FOR DATA SCIENCE

GROUP 4 PROJECT SUMMARY – STUDENTS' ACADEMIC PERFORMANCE

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ABSTRACT

What affects the academic performance of high-school students? This study examines the factors influencing student success, measured by Grade Point Average (GPA), using statistical and machine learning techniques, including Ordinary Least Squares (OLS) regression, propensity score matching, and Bayesian regression. Key findings show that absences significantly negatively affect GPA, while study time and parental support have positive effects. The analysis includes thorough validation using residual plots and metrics like R-squared and MSE. These findings highlight the importance of attendance and parental support in improving academic performance, offering valuable insights for educators and policymakers.

1 DATA SOURCE

This comprehensive dataset[1], sourced from Kaggle, includes information on the academic performance of 2,392 high school students. It covers demographics (ethnicity, gender, age), study habits (weekly study time, absences, tutoring), parental support and education, extracurricular activities, and academic performance (GPA and grade class). This dataset is particularly interesting because it provides insights into various factors that may influence student success, which can be valuable for educators and policymakers.

2 HYPOTHESIS AND OBJECTIVES

To investigate the relationship between GPA and independent variables, the following hypotheses were formulated:

Null Hypothesis (H_0):
There is no linear relationship between GPA and any independent variables:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \cdots = \beta_i = 0$$

Alternate Hypothesis (H_1):
At least one regression coefficient is not zero, indicating a significant correlation with GPA:

$$H_1: \ni \beta_0 \neq 0 \; for \; at \; least \; one \; i$$

The goal was to test these hypotheses and identify significant predictors of GPA, employing statistical regression models, causal inference, and Bayesian regression.

3 ANALYSIS – METHODS

Analysis involved data preprocessing, exploratory data analysis, one-way and two-way ANOVA, Ordinary Least Squares (OLS) regression, chi-square tests, t-tests, logistic regression, causal inference with propensity scores, and Bayesian regression. Specifically, one-way and two-way ANOVA tests were conducted to explore differences in mean GPA across categorical variables and their interactions. OLS regression, using both scaled and non-scaled models, was analyzed. Backward elimination method was used to remove non-significant variables based on p-values. Propensity score matching was performed to analyze the causal effect of absences on GPA using a multinomial logistic regression model and nearest neighbor matching. Bayesian regression helped in understanding the distribution of predicted GPA and assessing model assumptions.

4 ANALYSIS – RESULTS AND CONCLUSIONS

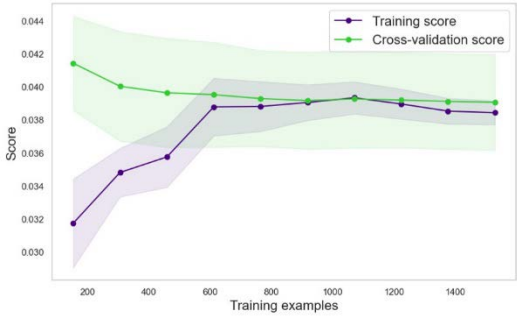
Minimal data preprocessing was required as the dataset did not contain null or duplicate values, and all

variables were assigned numerical values. ANOVA identified significant differences in GPA across various categorical and grouped continuous variables. Significant differences were found for tutoring, parental support, and extracurricular activities, while no significant differences were noted for gender, parental education, volunteering, and ethnicity. OLS regression was conducted to identify statistically significant linear relationships between GPA and independent variables. A backward elimination method was employed to remove non-significant variables, based on p-values. Both scaled and non-scaled versions of the model were analyzed to ensure robustness. Propensity score matching was performed to analyze the causal effect of the most significant predictor, absences, on GPA. Using a multinomial logistic regression model, nearest neighbor matching was performed to estimate treatment effects of different levels of absences on GPA. Finally, Bayesian regression helped in understanding the distribution of the predicted GPA and assessing the model's assumptions. The most significant predictors for academic performance were found to be absences, parental support, study time, tutoring, extracurricular activities, sports, and music participation. Gender, parental education, volunteering, and ethnicity did not show a significant direct effect on GPA. Most importantly, causal inference analysis indicated a strong negative impact of high absences on GPA, confirming the importance of regular attendance. According to all models, academic success is significantly affected by absences, parental support, and study time. Examining a wider student audience and adding more descriptive variables may help in better understanding what determines academic success.

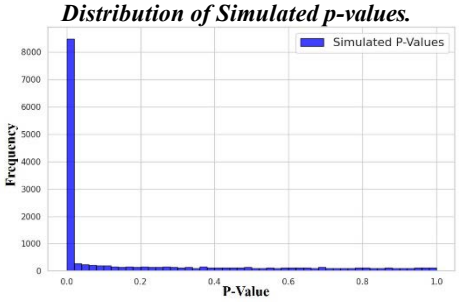
Variable Selection Using Backwards Elimination via OLS.

```
### Without Scaling ###
Removing variable Ethnicity_African_American with p-value 0.8543702843421989
Removing variable Ethnicity_Asian with p-value 0.9352347322351193
Removing variable Volunteering with p-value 0.6700012563695046
Removing variable Ethnicity_Caucasian with p-value 0.24220716502117942
Removing variable Gender with p-value 0.24256738184752644
Removing variable Age with p-value 0.16325362720837008
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GPA   R-squared:                       0.954
Model:                            OLS   Adj. R-squared:                  0.954
Method:                 Least Squares   F-statistic:                     5649.
Date:                Mon, 05 Aug 2024   Prob (F-statistic):               0.00
Time:                        13:16:38   Log-Likelihood:                 398.70
No. Observations:                1913   AIC:                            -781.4
Df Residuals:                    1905   BIC:                            -737.0
Df Model:                           7
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          2.5193      0.016    161.981      0.000       2.489       2.550
StudyTimeWeekly  0.0291      0.001     36.775      0.000       0.028       0.031
Absences      -0.0995      0.001   -187.392      0.000      -0.101      -0.098
Tutoring       0.2580      0.010     26.293      0.000       0.239       0.277
Extracurricular  0.1903      0.009     20.502      0.000       0.172       0.208
Sports         0.1856      0.010     18.965      0.000       0.166       0.205
Music          0.1521      0.011     13.455      0.000       0.130       0.174
ParentalSupport  0.1476      0.004     36.624      0.000       0.140       0.156
==============================================================================
Omnibus:                        5.236   Durbin-Watson:                   2.004
Prob(Omnibus):                  0.073   Jarque-Bera (JB):                4.971
Skew:                          -0.088   Prob(JB):                       0.0833
Kurtosis:                       2.824   Cond. No.                         70.1
==============================================================================
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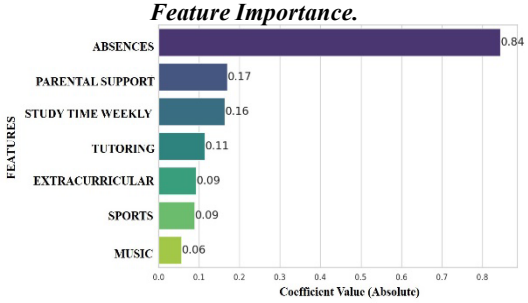
Learning Curves for Scaled OLS Regression.



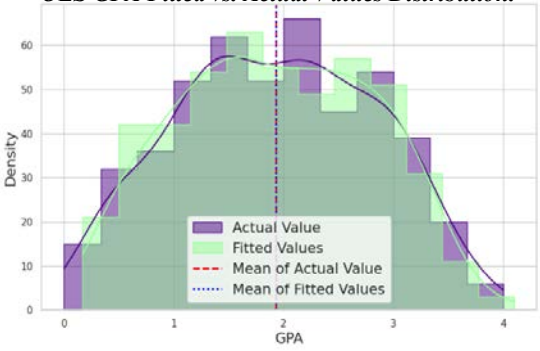
Distribution of Simulated p-values.



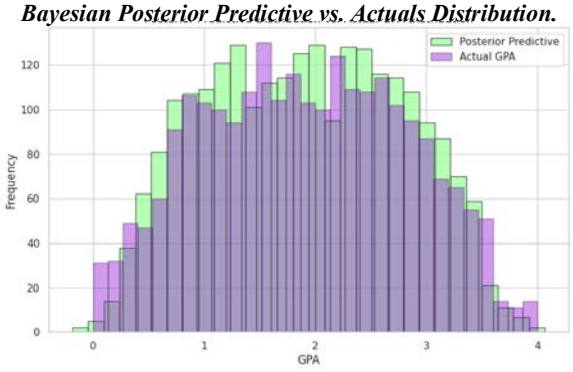
Feature Importance.



OLS GPA Fitted vs. Actual Values Distribution.



Bayesian Posterior Predictive vs. Actuals Distribution.



[1] Link: [students-performance-dataset/]